BSDS V: Regression Techniques

Problems Set 4: Model Selection

Consider the regression setup with response Y and a set of K predictors $\{x_1, \ldots, x_K\}$. Suppose, we want to evaluate all possible normal linear regression models that can be formed using a subset of these predictors and evaluate them. In particular, let N and F be the null (no predictor included) and full (all predictors included) models. In this context, answer the following questions:

1. For a model M, let $\bar{R}^2_M$ be the adjusted R^2 and RSS_M be the residual sum of squares under M. Show that
$$\frac{\mathrm{RSS}_M}{\mathrm{RSS}_F} = \frac{(n-K-1)}{(n-p_M-1)} \frac{(1-\bar{R}^2_M)}{(1-\bar{R}^2_F)},$$
where p_M is the number of predictors involved in the model M.

2. Consider the regression model
$$Y_i = \alpha + \gamma_1 x_i + \gamma_2 z_i + \varepsilon_i \qquad (i = 1, \ldots, n), \tag{*}$$
where
$$\sum_i x_i = \sum_i z_i = 0 \quad \text{and} \quad \sum_i x_i^2 = \sum_i z_i^2 = 1.$$
Suppose that we want to predict the response Y corresponding to a vector (x, z) using a predictor of the form $\hat{Y} = \overline{Y} + \tilde{\gamma}_1 x$, where $\tilde{\gamma}_1$ is the least squares estimate obtained by fitting the model
$$Y_i = \alpha + \gamma_1 x_i + \varepsilon_i \qquad (i = 1, \ldots, n).$$

(a) Assuming that the model (*) is correct, show that the expected model error
$$E[\mathrm{ME}] = E\left[(\hat{Y} - \alpha - \gamma_1 x - \gamma_2 z)^2\right]$$
of the predictor $\hat{Y}$ is given by
$$E[\mathrm{ME}] = \sigma^2(n^{-1} + x^2) + \gamma_2^2(rx - z)^2, \qquad \text{where} \quad r = \sum_i x_i z_i.$$

(b) Under what circumstances will $\hat{Y}$ be a better predictor than the predictor based on both x and z?